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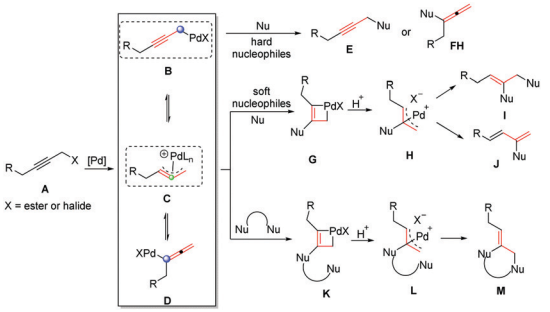
Palladium catalyzed divergent cycloadditions of vinylidenecyclopropane-diesters with methyleneindolinones enabled by zwitterionic π-propargyl palladium species†

Ben Niu,[a] Yin Wei[b] and Min Shi *[ab]

A palladium-catalyzed divergent synthesis of spirooxindoles fused with a five- or a six-membered ring by a cycloaddition reaction of vinylidenecyclopropane-diesters with methyleneindolinones was disclosed. This protocol features an *in situ* generated unprecedented zwitterionic π-propargyl palladium species in cycloaddition reactions and a switchable process between (3+2) and (4+2) cycloadditions by changing the phosphine ligand.

Organic synthesis based on propargylic compounds has received considerable attention and has been extensively studied due to their versatile and specific reactivities. Consequently, palladium-catalyzed reactions of propargylic compounds with nucleophiles have been extremely effective for the construction of carbon–carbon and carbon–heteroatom bonds.[1] The key step in these reactions was the formation of η^1-(propargyl)palladium species **B**, η^3-(propargyl)palladium species **C** or η^1-(allenyl)palladium species **D** from the propargylic donor **A** in the presence of palladium catalyst, which could undergo a variety of further transformations with suitable cross-partners (Scheme 1).[2] On the other hand, the regio- and chemoselectivities in the palladium-catalyzed propargyl substitution reaction are more complicated. According to the different reaction sites of nucleophiles and propargyl reagents, the branched products could be alkynes,[3] allenes,[4] alkenes[5] and conjugated dienes.[6] The previous investigations indicated that hard nucleophiles tended to attack the terminal or inner carbon atom of (propargyl)palladium species to afford alkyne products **E** or allene products **F**, while soft nucleophiles exhibited opposite reactivity to attack the central carbon atom of (propargyl)palladium species, affording alkene products **I** or conjugated diene products **J** (Scheme 1). Additionally, the palladium catalyzed annulation reactions of propargyl compounds and bis-nucleophiles have also been investigated.[7] The general process of cycloaddition reactions is described in Scheme 1. The first nucleophile attacks the central carbon atom of (propargyl)palladium species to deliver intermediate **K**, which can be converted to π-allyl palladium complex **L** upon protonation, then the second nucleophile attacks the π-allyl palladium intermediate to provide the cyclized products **M**. Obviously, the complex reaction properties of propargyl palladium greatly impede its application in synthetic chemistry. Moreover, the control of regio- and chemoselectivity is also a difficult task for its substitution reactions. With respect to the cyclization reaction, the substrates are restricted to those bearing dual soft nucleophilic functional groups, leading to a relatively single working mode. Therefore, it is quite significant to develop novel and efficient synthetic methods to effectively regulate regio- and chemoselective cyclization reactions, subsequently further enriching the reaction modes based on propargylic palladium species.

Vinylidenecyclopropanes (VDCPs), which have an allene moiety connected with a cyclopropane ring in their structural motifs, are a class of highly strained small ring compounds,



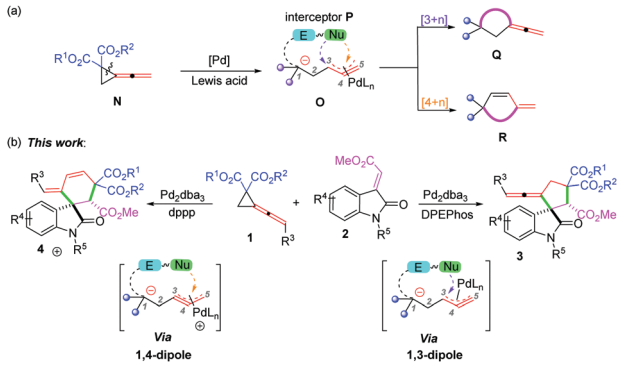
Scheme 1 The different reaction modes of propargyl palladium.

[a] *Key Laboratory for Advanced Materials and Institute of Fine Chemicals, School of Chemistry & Molecular Engineering, East China University of Science and Technology, 130 Mei Long Road, Shanghai 200237, People's Republic of China. E-mail: mshi@sioc.ac.cn*

[b] *State Key Laboratory of Organometallic Chemistry, Center for Excellence in Molecular Synthesis, University of Chinese Academy of Sciences, Shanghai Institute of Organic Chemistry, Chinese Academy of Sciences, 345 Lingling Road, Shanghai 200032, China*

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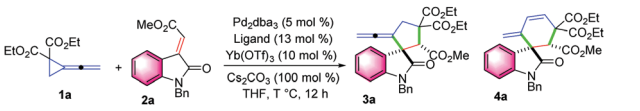


Scheme 2 Our working hypothesis and design of the divergent construction of spirooxindoles fused with a five- or six-membered ring.

which can easily undergo a variety of cyclopropane ring-opening reactions upon release of the strained ring energy.[8] As for vinylidenecyclopropane-diester **N** shown in Scheme 2, it incorporated a *gem*-diester in the cyclopropane moiety. Previously, our group and others have disclosed that a Lewis acid can effectively promote the cyclopropane ring-opening of vinylidenecyclopropane-diester **N** for a variety of interesting transformations.[9] Therefore, we envisioned that vinylidenecyclopropane-diester **N** could be used as a compatible π-propargyl donor to afford a zwitterionic π-propargyl palladium intermediate (**O**) in the presence of palladium catalyst with the assistance of a Lewis acid, which could act as a 1,3-(1*C*,3*C*) or 1,4-(1*C*,4*C*)-dipole to undergo cycloadditions with interceptor **P** to provide the corresponding (3+*n*) or (4+*n*) cyclization product **Q** or **R** (Scheme 2a). Herein, we wish to report our working design of the divergent construction of spirooxindoles fused with a five- or six-membered ring enabled by an *in situ* generated zwitterionic π-propargyl palladium species (Scheme 2b).[10]

Our initial examination was conducted with diethyl 2-vinylidenecyclopropane-1,1-dicarboxylate **1a** and methyl (*E*)-2-(1-benzyl-2-oxoindolin-3-ylidene)acetate **2a** in the presence of Pd_2dba_3 as a catalyst, XantPhos as a ligand, and $Yb(OTf)_3$ as an Lewis acid additive in tetrahydrofuran (THF) at 80 °C. Gratifyingly, the desired (3+2) cycloaddition product **3a** was obtained as a single diastereomer, albeit in a relatively low yield of 11% (Table 1, entry 1). When Cs_2CO_3 (1.0 eq.) was used as an additive, the yield of **3a** was improved to 36% presumably because the alkaline conditions are favorable for the formation of a zwitterionic π-propargyl palladium species (Table 1, entry 2). Subsequently, we examined a set of phosphine ligands and found that DPEPhos afforded **3a** in higher yield (42%) and *rac*-BINAP and XPhos failed to deliver the target product under identical conditions (Table 1, entries 3–5). Interestingly, when dppe or dppp was used as a ligand, the main product was switched to a (4+2) cycloadduct, **4a**, and the corresponding product could be obtained in 54% (dr: 5 : 1) and 53% (dr: 7 : 1) yields, respectively (Table 1, entries 6 and 7). Next, we employed DPEPhos as the ligand and performed this reaction at 60 °C; the desired product **3a** was achieved in 50% yield (Table 1, entry 8). An inferior result was obtained in the absence of $Yb(OTf)_3$ (Table 1, entry 9). Finally, we identified that by reducing the loadings of catalyst (2.5 mol%) and ligand (7.5 mol%) as well as using degassed THF as solvent, the corresponding product **3a** was obtained in 63% yield

Table 1 Optimization of the reaction conditions for Pd-catalyzed divergent cycloaddition reactions



Entry[a]	Ligand	*T*/°C	Yield[b] (%) 3a	4a (dr)
1[c]	XantPhos	80	11	
2	XantPhos	80	36	
3	*rac*-BINAP	80		Complex
4	XPhos	80		NR
5	DPEPhos	80	42	
6	dppe	80		54 (5 : 1)
7	dppp	80		53 (7 : 1)
8	DPEPhos	60	50	
9[d]	DPEPhos	60	32	
10[ef]	DPEPhos	60	63	
11	dppe	60		55 (4 : 1)
12	dppp	60		52 (8 : 1)
13	dppp	60		47 (9 : 1)
14[e]	dppp	80		56 (11 : 1)

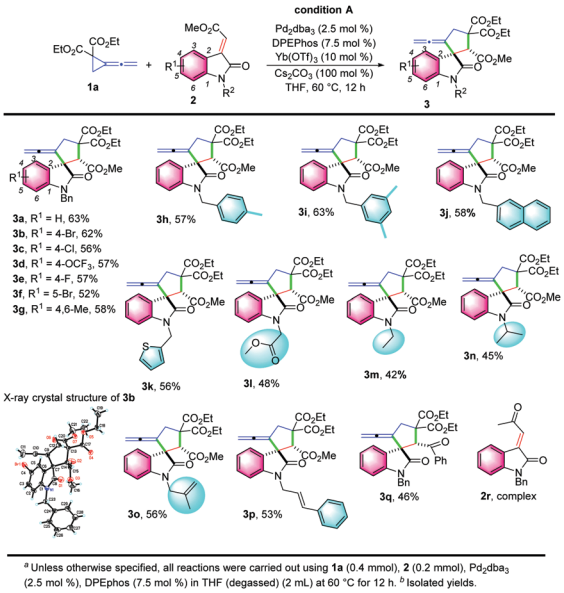
[a] The reaction was run under the following conditions: a solution of **1a** (0.4 mmol), **2a** (0.2 mmol), catalyst (5 mol%), and ligand (13 mol%) in dry solvent (2 mL). [b] Isolated yield. [c] The reaction was performed without Cs_2CO_3. [d] The reaction was performed without $Yb(OTf)_3$. [e] Pd_2dba_3 (2.5 mol%), ligand (7.5 mol%). [f] Degassed THF.

(Table 1, entry 10, conditions **A**). Next, we turned our attention to further optimize the reaction conditions for **4a**. When the reactions were carried out at 60 °C, the use of dppe, dppp and dppb did not give better results (Table 1, entries 11–13). We also found that the yield of **4a** could be improved to 56% with an 11 : 1 dr by reducing the loadings of catalyst (2.5 mol%) and ligand (7.5 mol%) as well as performing the reaction at 80 °C (Table 1, entry 14, conditions **B**) (for more details see Schemes S1 and S2 in the ESI†). We also investigated the stability of products **3a** and **4a** under conditions **A** and conditions **B**, respectively, and found that **3a** and **4a** are stable under their respective optimal conditions.

On the basis of the above optimized conditions for this divergent cycloaddition reaction, the scope of (3+2) cycloaddition reactions between vinylidenecyclopropane-diester **1a** and methyleneindolinones **2** was first explored and the results are shown in Scheme 3. Fixing the R^2 protecting group as a benzyl group, the electronic effects at the different positions of the benzene ring of isatins were investigated. We found that regardless of whether the substituent was introduced at the C4-position (**2b and e**) or C5-position (**2f**), the reactions proceeded smoothly, furnishing the desired products **3b–3f** in 52–62% yields. Substrate **2g** bearing 4,6-dimethyl groups at the benzene ring of isatin was also tolerated, providing the desired product **3g** in 58% yield. Substrate **2h** bearing the 4-methyl group at the benzyl protecting group and substrate **2i** having 3,5-dimethyl groups at the benzyl protecting group were also compatible, affording the desired products **3h** and **3i** in 57% and 63% yields, respectively. Then, we turned our attention towards the different types of nitrogen-protecting groups. For substrates **2j–2p**, in which the nitrogen atom was protected

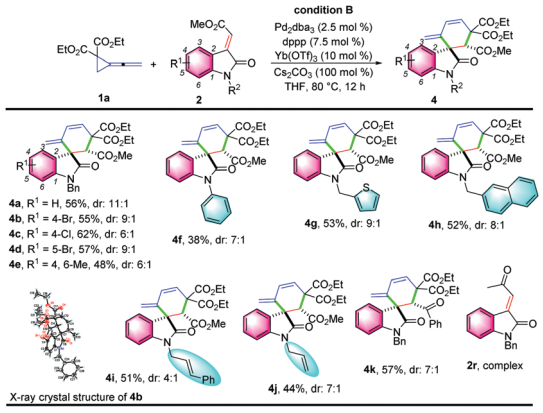
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[a] Unless otherwise specified, all reactions were carried out using **1a** (0.4 mmol), **2** (0.2 mmol), Pd_2dba_3 (2.5 mol %), DPEphos (7.5 mol %) in THF (degassed) (2 mL) at 60 °C for 12 h. [b] Isolated yields.

Scheme 3 Substrate scope for the (3+2) cycloaddition reaction[a,b].



[a] Unless otherwise specified, all reactions were carried out using **1a** (0.4 mmol), **2** (0.2 mmol), Pd_2dba_3 (2.5 mol %), dppp (7.5 mol %) in THF (degassed) (2 mL) at 80 oC for 12 h. [b] Isolated yields.

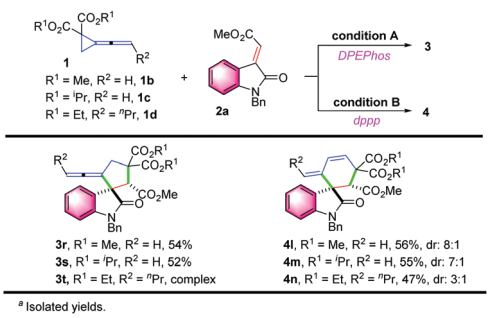
Scheme 4 Substrate scope for the (4+2) cycloaddition reaction[a,b].

with 2-methylnaphthalene, 2-methylthiophene, methylacetate, ethyl, isopropyl, 2-methylallyl and cinnamyl groups, the reactions proceeded smoothly, giving the desired products **3j–3p** in moderate yields ranging from 42 to 58%. Under the same reaction conditions, (*E*)-1-benzyl-3-(2-oxo-2-phenylethylidene) indolin-2-one **2q** could also be converted to the corresponding product **3q** in 46% yield. However, when (*E*)-1-benzyl-3-(2-oxo propylidene)indolin-2-one **2r** was employed as the substrate, a complex product mixture was formed, presumably due to its instability in the presence of Cs_2CO_3. The structure of **3b** was unambiguously determined by X-ray diffraction, and its ORTEP drawing is shown in Scheme 3.

Subsequently, the scope of (4+2) cycloaddition reaction was also investigated under the optimal conditions (Scheme 4). Substrates **2** bearing different substituents at the C4-position or C5-position could be converted to the corresponding products **4b–4d** in 55–62% yields with 6 : 1–11 : 1 dr values. The substrate bearing 4,6-dimethyl groups at the benzene ring of isatin was also tolerated, providing the desired product **4e** in 48% yield with a 6 : 1 dr. Upon replacing the benzyl protecting group with a phenyl group, the reaction could also take place, affording the desired product **4f** in 38% yield with a 7 : 1 dr. When the nitrogen atom was protected with different protecting groups, such as 2-methylnaphthalene, 2-methylthiophene, cinnamyl and allyl groups, the desired products **4g–4j** could be obtained in 44–53% yield with 4 : 1–9 : 1 dr values. As mentioned above, the corresponding product **4k** could also be acquired in 57% yield with a 7 : 1 dr when using (*E*)-1-benzyl-3-(2-oxo-2-phenylethylidene)indolin-2-one **2q** as a substrate. However, **2r** was also incompatible with the optimized conditions. The X-ray crystal structure of **4b** is shown in Scheme 4.

The above developed optimal conditions were also applied to a variety of vinylidenecyclopropane-diesters **1**. As summarized in Scheme 5, vinylidenecyclopropanes substituted with methyl or iso-propyl esters could also work under both conditions **A** and **B**, delivering the corresponding products **3r and 3s** and **4l and 4m** (dr: 7 : 1–8 : 1) in moderate yields. When *N*-propyl was introduced at the terminal allene moiety of **1**, the (4+2) cycloaddition product **4n** was obtained in 47% yield with a 3 : 1 dr under conditions **B**, but a complex mixture was produced under conditions **A**.

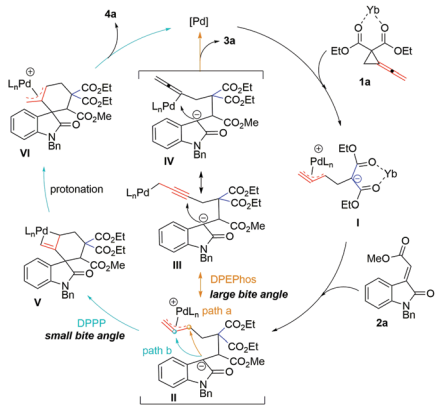
A plausible mechanism for the (3+2) and (4+2) cycloadditions is depicted in Scheme 6. The catalytic cycle was initiated by the Pd–π-propargyl species **I**, formed from Pd(0) and **1a** with the assistance of Lewis acid $Yb(OTf)_3$. The nucleophilic carbon anion of intermediate **I** attacked **2a** to afford intermediate **II**. On one hand, by using DPEPhos having a large bite angle as a ligand,[11] the carbon anion selectively attacked the inner carbon atom to afford the (3+2) cycloaddition product **3a** *via* its resonance structure **III** or **IV** (path a). On the other hand, using dppp having a small bite angle as a ligand, the carbon anion selectively attacked the Pd–π-propargyl unit at its central carbon atom to generate the corresponding pallada-cyclobutene intermediate **V**, which could produce the π-allyl palladium species **VI** upon protonation. The subsequent base-induced β-hydride[12] elimination would regenerate the palladium(0) catalyst and produce the corresponding product **4a** (path b). A more detailed mechanistic investigation will be carried out in due course.



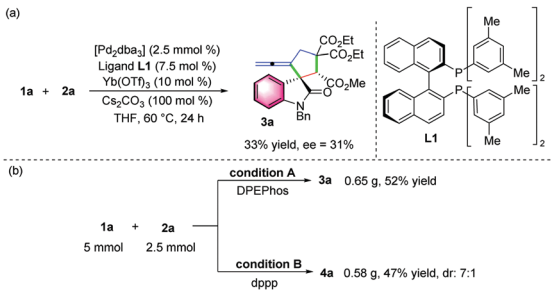
[a] Isolated yields.

Scheme 5 Scope of vinylidenecyclopropane-diesters in reactions with (*E*)-2-(1-benzyl-2-oxoindolin-3-ylidene)acetate **2a**[a].

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Scheme 6 Proposed reaction mechanism.



Scheme 7 Preliminary asymmetric studies and a scale-up synthesis.

The catalytic asymmetric version of this divergent cycloaddition was also investigated (see Table S1 in the ESI† for more details) and we found that using **L1** as a chiral phosphine ligand could deliver **3a** in 33% yield along with 31% ee (Scheme 7a). Finally, a scale-up synthesis was performed with 5.0 mmol of substrate **1a** and 2.5 mmol of substrate **2a** under both conditions **A** and **B**, affording the corresponding products **3a** and **4a** in 52% and 47% (dr: 7 : 1) yields, respectively (Scheme 7b).

In summary, we have disclosed an efficient and convenient synthetic protocol to produce spirooxindoles fused with a five or six-membered ring by the reaction of vinylidenecyclopropanediesters with methyleneindolinones in the presence of palladium catalyst and a Lewis acid. By simply switching the ligands, an unprecedented zwitterionic π-propargyl palladium catalyzed regiodivergent cyclization is realized to enable (3+2) and (4+2) cycloadditions. This discovery sheds light on the development of propargyl palladium species related reactions and enriches the diversity of its reaction mode. Further investigations on the asymmetric version of this palladium-catalyzed divergent cycloaddition and application of this protocol to the synthesis of biologically active substances are underway.

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Conflicts of interest

There are no conflicts of interest to declare.

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